What you did at home, round two

AI Methods for Science – Part I – Week 3, Monday

E9–E15: **Nadav** Am Shalom **Oran** Szachter **David** Shem-Tov

E0–E8: **Alexander** Leibenzon **Ido** Shachal

26 October 2026

1. The default run, on eight machines

`val_acc_at_best`, default config, 20 epochs

seed	Yonatan	Amit & Adi	Oran	Tamar	Alexander	Ido	Nadav	David
0	51.8	51.8	**54.1**	53.5	53.3	53.6	53.5	53.0
1	52.8	52.8	52.8	52.5	52.4	52.0	52.9	52.3
2	51.6	51.6	51.6	50.9	**51.0**	51.6	52.1	51.3

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The baseline you compare everything to is 51.8–54.1, depending on the machine. A gain of less than 2 points is not a gain.

2. Weight decay

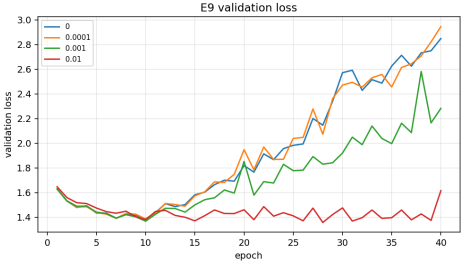


figure: **Nadav**. E9, validation loss.

`val_acc_at_best`

λ	0	10^{-4}	10^{-3}	10^{-2}
ours	51.8	51.6	53.3	53.7
Nadav	53.5	53.4	53.6	53.5
Oran	54.1	53.5	53.2	53.4
David	53.0	54.3	54.1	53.2
train acc, 60 ep (ours)	100	100	99	**64**

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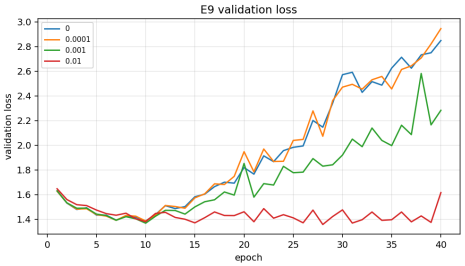


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Here it delays the loss rise and lowers the training accuracy; the validation accuracy moves inside the noise. *This network, with early stopping*: not a rule. On a CNN it is worth points.

3. Dropout

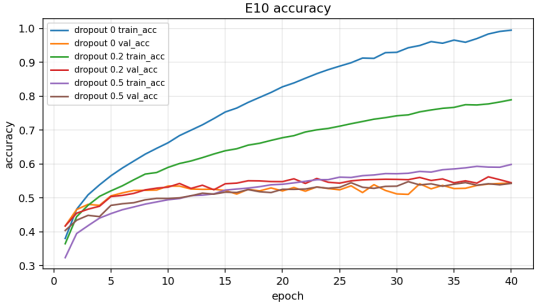


figure: **Nadav**. E10, training and validation accuracy.

`val_acc_at_best`, and the gain over the baseline

dropout	0	0.2		0.5	
ours	51.8	55.0	+3.2	55.4	+3.6
Nadav	53.5	55.7	+2.2	54.5	+1.0
Oran	54.1	54.9	+0.8	54.8	+0.7
David	53.0	54.1	+1.1	55.1	+2.1

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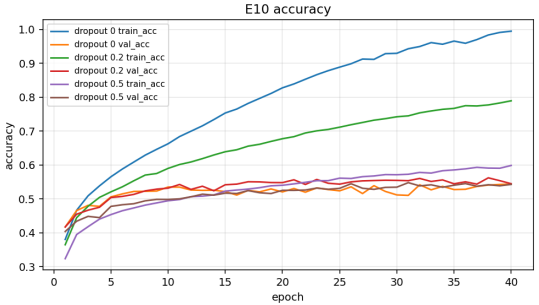


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Dropout lands at 54–56 on every machine; the gain is whatever your baseline happened to be.

4. Optimizers

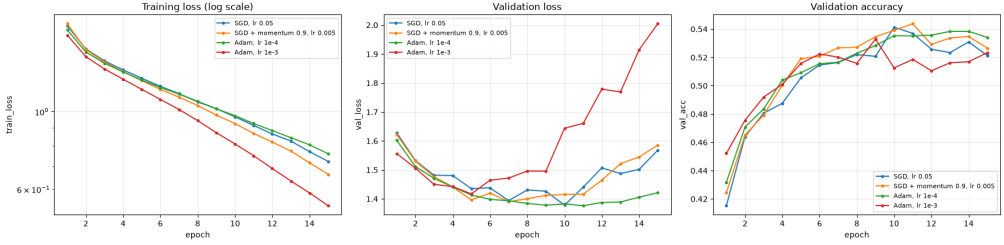


figure: **Oran**. E12, 15 epochs.

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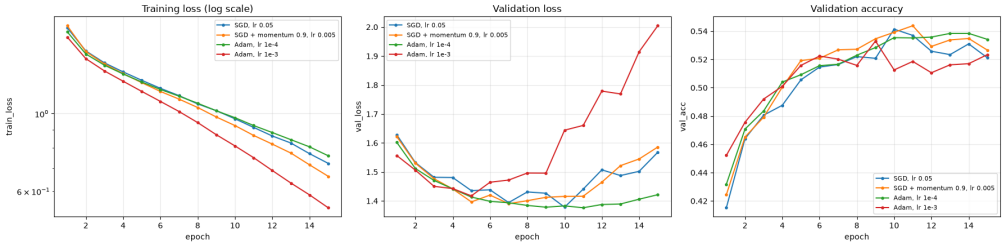


figure: **Oran**. E12, 15 epochs.

Adam at 10^{-3} gets the lowest training loss and overfits first. All four end within 2 points of each other.

5. Eight layers

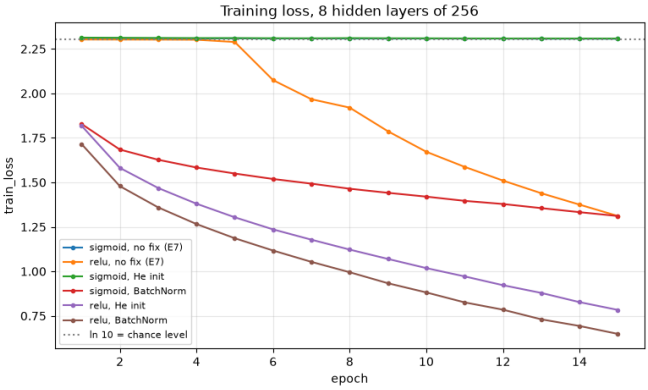


figure: **Oran**. E13, training loss, [256]∗8.

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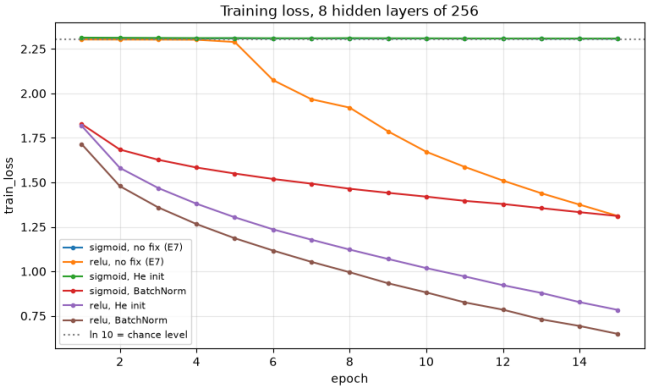


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He init fixes ReLU, not sigmoid: $\sigma' \leq \frac{1}{4}$ is a property of σ, not of the scale of W. BatchNorm fixes both.

6. Augmentation

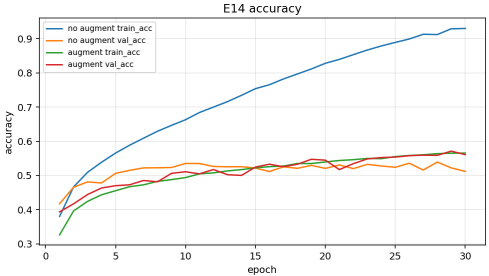


figure: **Nadav**. E14, training and validation accuracy.

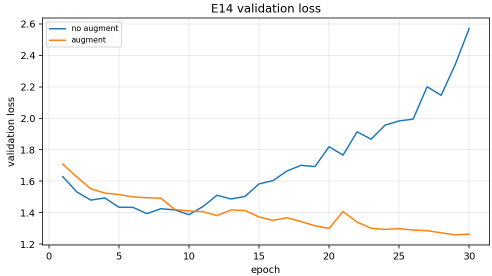


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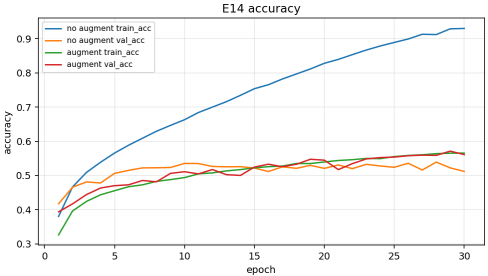


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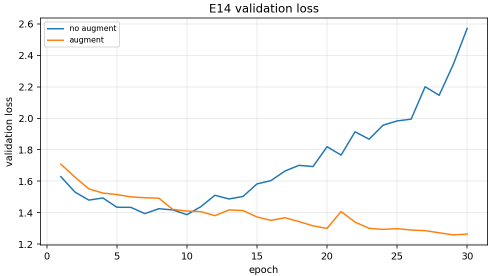


figure: **Nadav**. E14, validation loss.

The validation loss never turns up: 1.26 at epoch 30 on every machine. $+4$ points, more than any penalty on θ.

7. Everything together

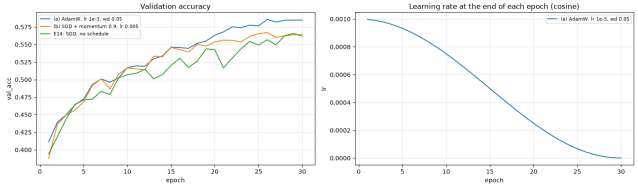


figure: **Oran**. E15, validation accuracy and the AdamW learning rate.

AdamW, cosine, augment, test once

	val	test
ours, 30 ep	58.1	58.0
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David, 30 ep	58.0	58.4
Nadav, **20 ep**	55.5	56.4

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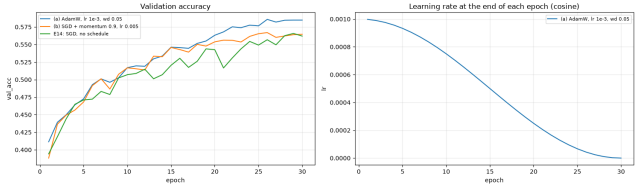


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Best epoch = last, test = validation, 58. Same network as E11 (54.6); it still does not know that neighbouring pixels are neighbours.